

Yohan Shanuka Jayakody

Machine Learning | MLOps Engineer

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

PROFILE

Final-year BICT (Hons.) undergraduate with hands-on experience in **Machine Learning**, **Agentic AI**, and **MLOps**. Skilled in developing intelligent AI/ML systems, predictive models, autonomous AI workflows, and scalable data pipelines using Python, SQL, Spark, Kafka, Airflow, Docker, AWS, and MLflow.

TECHNICAL SKILLS

Programming – Python, SQL

Machine Learning - Scikit-learn, TensorFlow, Keras, XGBoost, CNN, Pandas, NumPy, Matplotlib

Agentic AI– LangChain, LangFuse, LangGraph, HuggingFace, Supabase, Qdrant

Data Engineering – Apache Spark, Airflow, Kafka, NIFI, ETL, Data Warehousing

Databases – PostgreSQL, MySQL, MongoDB

Cloud & DevOps – Docker, Git, GitHub, AWS (S3, EC2, RDS, ECS), MLflow, DVC

BI & Visualization – Amazon QuickSight, Grafana

FEATURED PROJECTS

End-to-End Customer Churn Prediction Pipeline

Tech Stack: Python, Scikit-learn, Spark, Kafka, Airflow, PostgreSQL, MLflow, Docker, AWS QuickSight (GitHub:

github.com/yshanukajay/churn-pipeline-deployment-last)

- Built an end-to-end customer churn prediction pipeline with scalable ETL workflows, automated data processing, ML model training and evaluation, MLflow experiment tracking, Docker containerization, and Amazon QuickSight dashboards for actionable insights.

Real-Time Credit Card Fraud Detection System

Tech Stack: Python, XGBoost, Kafka, Spark, Airflow, PostgreSQL, MLflow, Docker, AWS (S3, ECS/ECR) (GitHub:

github.com/yshanukajay/end-to-end-credit-card-fraud-detection-system)

- Built an end-to-end real-time fraud detection pipeline using Kafka, Spark, and Airflow, featuring an optimized XGBoost model, MLflow experiment tracking, Docker containerization, AWS deployment, and Grafana dashboards for real-time monitoring and insights.

RESEARCH & PUBLICATIONS

Real-Time Cattle Monitoring Using Low-Cost IoT Smart Collars with LoRa Communication in Sri Lanka's Dry Zones (RCAICT - 2025) - [Visit](#)

- Co-authored a research paper on a low-cost IoT-based smart collar system for real-time cattle monitoring using GPS, environmental sensors, and LoRa communication.
- Evaluated the system through field experiments, demonstrating reliable LoRa communication over approximately **3 km** in rural dry-zone environments.

EDUCATION

Bachelor of Information & Communication Technology(Hons.)

2023-Present

Faculty of Technological Studies, University of Vavuniya, Sri Lanka

CERTIFICATIONS

IBM Machine Learning Specialization - IBM ([Link](#))

Machine Learning Specialization - DeepLearning.AI ([Link](#))

Generative AI with Large Language Models – AWS ([Link](#))